

Jérémie Dentan

*PhD candidate in Machine Learning
Security and Privacy*

✉ jeremie.dentan@polytechnique.org
📄 <https://dentanjeremie.github.io>
Paris, France

About me

I am currently pursuing a PhD in [machine learning security and privacy](#) at École Polytechnique, Paris. My research focuses on emerging challenges in large language models, including data memorization, uncertainty quantification and adversarial robustness. Expected date of defense: December 2026.

Selected Publications

- Preprint *Kernel Token Contradiction: a Fast and Principled Approach for LLM Claim Uncertainty Quantification.*
Jérémie Dentan, Alexi Canesse, Mohammed El Sharkawy, Sonia Vanier.
<https://arxiv.org/abs/2608.22506>
- AAAI-26 *Guess or Recall? Training CNNs to Classify and Localize Memorization in LLMs.*
(oral) **Jérémie Dentan**, Davide Buscaldi, Sonia Vanier.
<https://doi.org/10.1609/aaai.v40i36.40304>
- LREC-26 *MUCH: A Multilingual Claim Hallucination Benchmark.*
Jérémie Dentan, Alexi Canesse, Davide Buscaldi, Aymen Shabou, Sonia Vanier.
<https://doi.org/10.63317/4zrfhra4azck>
- ECAI 2025 *Predicting memorization within Large Language Models fine-tuned for classification.*
(spotlight) **Jérémie Dentan**, Davide Buscaldi, Aymen Shabou Sonia Vanier.
<https://doi.org/10.3233/FAIA251323>
- Usenix Security 24 *Reconstructing Training Data From Document Understanding Models.*
Jérémie Dentan, Arnaud Paran, Aymen Shabou.
<https://www.usenix.org/conference/usenixsecurity24/presentation/dentan>

Work experience

- Mar 2026– **AI Scientist (Intern), Mistral AI** , Paris, France .
- Aug 2026 [Post-training team for Mistral Large 4](#). Brought new abilities to the model in code agent setting. Introduced a new algorithm to improve supervised fine-tuning speed by [9-12%](#).
- Apr 2023– **Data Scientist (Intern), Crédit Agricole DataLab Group** , Paris, France .
- Sept 2023 Worked on the first [model inversion attacks](#) that target document processing models. Worked published at Usenix Security 2024.
- Apr 2022– **Software Developer (Intern), Oracle Labs** , Zürich, Switzerland .
- Aug 2022 Worked on an internal tool for risk prediction, with a focus on its [entity resolution](#) pipeline.
- Jun 2021– **Business Analyst (Intern), Public Sector team, BearingPoint** , Paris, France .
- Aug 2021 [Digital transformation of administrations](#): project scope; funding plan; project owner support.
- Jan 2021– **Project Manager and Quality Manager, XProjets** , Palaiseau, France .
- Dec 2021 A small consulting firm (100–150k€ turnover) run by a dozen students of Polytechnique.
[Main accomplishment in 2021](#): XProjets ranked in L30 (top 30 junior companies in France).
- Nov 2019– **Deputy Project Manager, French Embassy** , Antananarivo, Madagascar .
- Mar 2020 French armed-forces officer in the diplomatic community, working on the French-Malagasy cooperation for civil protection.

Education

- Nov 2023– **PhD Candidate, École Polytechnique** , *Palaiseau, France* .
- Nov 2026 Measuring and Improving the Trustworthiness of Language Models.
- What amount of training data can we extract from language model?
 - How to audit models during training and protect samples from memorization?
 - Why does memorization happens and where is it encoded in the LLM?
- Sep 2022– **Master MVA: "Mathematics, Vision, Learning", ENS Paris-Saclay** , *France* .
- Aug 2023 A one year graduating program to specialize in machine learning and applied mathematics.
- ML courses: Convex Optimization, Deep Learning, Kernel Methods.
 - Theoretical courses: Model & Information Theory, Geometric Data Analysis.
 - Applications courses: ML for time series, ML for text and graph data, Detection theory.
 - Seminars: Reproducible research and free software, Responsible ML.
- [Final grade](#): 17.86/20 (highest honours).
- [Master thesis](#): *Reconstructing Training Data from Document Understanding Models*.
- Sep 2019– **Master of Engineering, École Polytechnique Paris** , *France* .
- Aug 2022 "*Cycle ingénieur polytechnicien, promotion X2019*"
- [Computer Science and Machine Learning](#): Algorithmics, Deep Learning, Reinforcement Learning, Systems for Big Data, NLP, Database Management, Data Visualization, Federated Learning, Web Development, Object-oriented programming, Theoretical Computer Science.
 - [Applied Mathematics](#): Statistics, Optimization and Control, Statistics.
 - [Business](#): Microeconomics, Macroeconomics, Corporate Governance and CSR.
- [Cumulative GPA](#): 3.89 (top 20%).
- Sep 2017– **"Classe Préparatoire MPSI/MP*", Sainte Geneviève High School** , *France* .
- Aug 2019 Similar to the beginning of a BSc in Mathematics and Physics, it is a 2-year intensive program preparing for the national competitive exam for entry to engineering schools.
- [Cumulative GPA](#): 3.99 (ranked 5 out of 150).
- Jun 2017 **French Baccalaureate, Military High School of Aix-en-Provence** , *France* .
- Mathematics: 18th at "Concours Général" (most selective A-level competition in France).
- [Ranking](#): 1st of Aix-Marseille Academy (>25k students). Grade: 20.68/20.

Languages and IT

- Languages [French](#) (native), [English](#) (fluent, TOEFL 100/120), German (elementary level, B1 certified).
- IT Languages [Preferred language: Python](#). Knowledges in JS, HTML, Java, C++, D3, VBA.

Interests and activities

- Passions Very strong interest for classical and romantic music, played the piano for 12 years.
- Sports Running, climbing, and mountaineering. I have also practiced tennis, judo, and badminton.
- Army Merit Medal of the Malagasy Army, 2020, for my work for the military cooperation.